6 THE NORMAL DISTRIBUTION



Figure 6.1 If you ask enough people about their shoe size, you will find that your graphed data is shaped like a bell curve and can be described as normally distributed. (credit: modification of work "Shoe shop" by Lo/ Flickr, Public Domain)

Chapter Objectives

By the end of this chapter, the student should be able to:

- Recognize the normal probability distribution and apply it appropriately.
- Recognize the standard normal probability distribution and apply it appropriately.
- Compare normal probabilities by converting to the standard normal distribution.

Introduction

The normal, a continuous distribution, is the most important of all the distributions. It is widely used and even more widely abused. Its graph is bell-shaped. You see the bell curve in almost all disciplines. Some of these include psychology, business, economics, the sciences, nursing, and, of course, mathematics. Some of your instructors may use the normal distribution to help determine your grade. Most IQ scores are normally distributed. Often real-estate prices fit a normal distribution. The normal distribution is extremely important, but it cannot be applied to everything in the real world.

In this chapter, you will study the normal distribution, the standard normal distribution, and applications associated with them.

The normal distribution has two parameters (two numerical descriptive measures): the mean (μ) and the standard deviation (σ). If X is a quantity to be measured that has a normal distribution with mean (μ) and standard deviation (σ), we designate this by writing

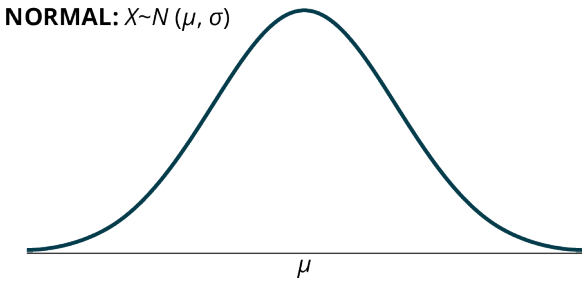


Figure 6.2

The probability density function is a rather complicated mathematical function. **Do not memorize it**. It is not necessary.

$$f(x) = \frac{1}{\sigma \cdot \sqrt{2 \cdot \pi}} \cdot e^{-\frac{1}{2} \cdot \left(\frac{x-\mu}{\sigma}\right)^2}$$

The cumulative distribution function is $P(X < x)$. It is calculated either by a calculator or a computer, or it is looked up in a table. Technology has made the tables virtually obsolete. For that reason, as well as the fact that there are various table formats, we are not including table instructions.

The curve is symmetric about a vertical line drawn through the mean, μ. In theory, the mean is the same as the median, because the graph is symmetric about μ. As the notation indicates, the normal distribution depends only on the mean and the standard deviation. Since the area under the curve must equal one, a change in the standard deviation, σ, causes a change in the shape of the curve; the curve becomes fatter or skinnier depending on σ. A change in μ causes the graph to shift to the left or right. This means there are an infinite number of normal probability distributions. One of special interest is called the **standard normal distribution**.

COLLABORATIVE EXERCISE

Your instructor will record the heights of both men and women in your class, separately. Draw histograms of your data. Then draw a smooth curve through each histogram. Is each curve somewhat bell-shaped? Do you think that if you had recorded 200 data values for men and 200 for women that the curves would look bell-shaped? Calculate the mean for each data set. Write the means on the x-axis of the appropriate graph below the peak. Shade the approximate area that represents the probability that one randomly chosen male is taller than 72 inches. Shade the approximate area that represents the probability that one randomly chosen female is shorter than 60 inches. If the total area under each curve is one, does either probability appear to be more than 0.5?

6.1 The Standard Normal Distribution

The **standard normal distribution** is a normal distribution of **standardized values called *z*-scores**. **A *z*-score is measured in units of the standard deviation.** For example, if the mean of a normal distribution is five and the standard deviation is two, the value 11 is three standard deviations above (or to the right of) the mean. The calculation is as follows:

$x = \mu + (z)(\sigma) = 5 + (3)(2) = 11$

The z-score is three.

The mean for the standard normal distribution is zero, and the standard deviation is one. The transformation $z = \frac{x-\mu}{\sigma}$ produces the distribution $Z \sim N(0, 1)$. The value x in the given equation comes from a normal distribution with mean μ and standard deviation σ.

Z-Scores

If X is a normally distributed random variable and $X \sim N(\mu, \sigma)$, then the z-score is:

$$z = \frac{x - \mu}{\sigma}$$

The *z*-score tells you how many standard deviations the value *x* is above (to the right of) or below (to the left of) the mean, *μ*. Values of x that are larger than the mean have positive z-scores, and values of x that are smaller than the mean have negative z-scores. If x equals the mean, then x has a z-score of zero.

EXAMPLE 6.1

Suppose $X \sim N(5, 6)$. This says that X is a normally distributed random variable with mean $\mu = 5$ and standard deviation $\sigma = 6$. Suppose $x = 17$. Then:

$$z = \frac{x-\mu}{\sigma} = \frac{17-5}{6} = 2$$

This means that $x = 17$ is **two standard deviations** (2σ) above or to the right of the mean $\mu = 5$.

Notice that: $5 + (2)(6) = 17$ (The pattern is $\mu + z\sigma = x$)

Now suppose $x = 1$. Then: $z = \frac{x-\mu}{\sigma} = \frac{1-5}{6} = -0.67$ (rounded to two decimal places)

This means that $x = 1$ is 0.67 standard deviations (-0.67σ) below or to the left of the mean $\mu = 5$. Notice that: $5 + (-0.67)(6)$ is approximately equal to one (This has the pattern $\mu + (-0.67)\sigma = 1$)

Summarizing, when z is positive, x is above or to the right of μ and when z is negative, x is to the left of or below μ. Or, when z is positive, x is greater than μ, and when z is negative x is less than μ.

TRY IT 6.1

What is the z-score of x, when $x = 1$ and $X \sim N(12,3)$?

EXAMPLE 6.2

Some doctors believe that a person can lose five pounds, on the average, in a month by reducing their fat intake and by exercising consistently. Suppose weight loss has a normal distribution. Let X = the amount of weight lost (in pounds) by a person in a month. Use a standard deviation of two pounds. $X \sim N(5, 2)$. Fill in the blanks.

Problem

a. Suppose a person **lost** ten pounds in a month. The z-score when $x = 10$ pounds is $z = 2.5$ (verify). This z-score tells you that $x = 10$ is _____ standard deviations to the _____ (right or left) of the mean ____ (What is the mean?).

Solution

a. This z-score tells you that $x = 10$ is **2.5** standard deviations to the **right** of the mean **five**.

Problem

b. Suppose a person **gained** three pounds (a negative weight loss). Then $z =$ _____. This z-score tells you that $x = -3$ is _____ standard deviations to the _____ (right or left) of the mean.

Solution

b. $z =$ **–4**. This z-score tells you that $x = -3$ is **four** standard deviations to the **left** of the mean.

Problem

c. Suppose the random variables X and Y have the following normal distributions: $X \sim N(5, 6)$ and $Y \sim N(2, 1)$. If $x = 17$, then $z = 2$. (This was previously shown.) If $y = 4$, what is z?

Solution

c. $z = \frac{y-\mu}{\sigma} = \frac{4-2}{1} = 2$ where $\mu = 2$ and $\sigma = 1$.

The z-score for $y = 4$ is $z = 2$. This means that four is $z = 2$ standard deviations to the right of the mean. Therefore, $x = 17$ and $y = 4$ are both two (of **their own**) standard deviations to the right of **their** respective means.

The z-score allows us to compare data that are scaled differently. To understand the concept, suppose $X \sim N(5, 6)$ represents weight gains for one group of people who are trying to gain weight in a six week period and $Y \sim N(2, 1)$ measures the same weight gain for a second group of people. A negative weight gain would be a weight loss. Since $x = 17$ and $y = 4$ are each two standard deviations to the right of their means, they represent the same, standardized weight gain **relative to their means**.

TRY IT 6.2

Fill in the blanks.

Jerome averages 16 points a game with a standard deviation of four points. $X \sim N(16,4)$. Suppose Jerome scores ten points in a game. The z–score when $x = 10$ is –1.5. This score tells you that $x = 10$ is _____ standard deviations to the _____(right or left) of the mean_____(What is the mean?).

The Empirical Rule

If X is a random variable and has a normal distribution with mean μ and standard deviation σ, then the **Empirical Rule** states the following:

- About 68% of the x values lie between -1σ and $+1\sigma$ of the mean μ (within one standard deviation of the mean).
- About 95% of the x values lie between -2σ and $+2\sigma$ of the mean μ (within two standard deviations of the mean).
- About 99.7% of the x values lie between -3σ and $+3\sigma$ of the mean μ (within three standard deviations of the mean). Notice that almost all the x values lie within three standard deviations of the mean.
- The z-scores for $+1\sigma$ and -1σ are +1 and –1, respectively.
- The z-scores for $+2\sigma$ and -2σ are +2 and –2, respectively.
- The z-scores for $+3\sigma$ and -3σ are +3 and –3 respectively.

The empirical rule is also known as the 68-95-99.7 rule.

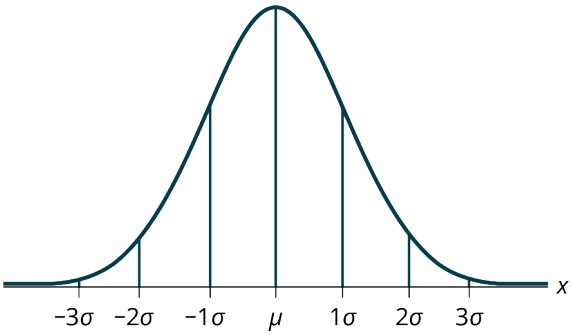


Figure 6.3

EXAMPLE 6.3

The mean height of 15 to 18-year-old males from Chile from 2009 to 2010 was 170 cm with a standard deviation of 6.28 cm. Male heights are known to follow a normal distribution. Let X = the height of a 15 to 18-year-old male from Chile in 2009 to 2010. Then $X \sim N(170, 6.28)$.

Problem

a. Suppose a 15 to 18-year-old male from Chile was 168 cm tall from 2009 to 2010. The z-score when $x = 168$ cm is $z =$ _____. This z-score tells you that $x = 168$ is _____ standard deviations to the _____ (right or left) of the mean _____ (What is the mean?).

b. Suppose that the height of a 15 to 18-year-old male from Chile from 2009 to 2010 has a z-score of $z = 1.27$. What is the male's height? The z-score ($z = 1.27$) tells you that the male's height is _____ standard deviations to the _____ (right or left) of the mean.

Solution

a. –0.32, 0.32, left, 170

b. 177.98 cm, 1.27, right

TRY IT 6.3

Use the information in Example 6.3 to answer the following questions.

a. Suppose a 15 to 18-year-old male from Chile was 176 cm tall from 2009 to 2010. The *z*-score when $x = 176$ cm is $z =$ _____. This *z*-score tells you that $x = 176$ cm is _____ standard deviations to the _____ (right or left) of the mean _____ (What is the mean?).
b. Suppose that the height of a 15 to 18-year-old male from Chile from 2009 to 2010 has a *z*-score of $z = -2$. What is the male's height? The *z*-score ($z = -2$) tells you that the male's height is _____ standard deviations to the _____ (right or left) of the mean.

EXAMPLE 6.4

Problem

From 1984 to 1985, the mean height of 15 to 18-year-old males from Chile was 172.36 cm, and the standard deviation was 6.34 cm. Let Y = the height of 15 to 18-year-old males from 1984 to 1985. Then $Y \sim N(172.36, 6.34)$.

The mean height of 15 to 18-year-old males from Chile from 2009 to 2010 was 170 cm with a standard deviation of 6.28 cm. Male heights are known to follow a normal distribution. Let X = the height of a 15 to 18-year-old male from Chile in 2009 to 2010. Then $X \sim N(170, 6.28)$.

Find the *z*-scores for $x = 160.58$ cm and $y = 162.85$ cm. Interpret each *z*-score. What can you say about $x = 160.58$ cm and $y = 162.85$ cm as they compare to their respective means and standard deviations?

Solution

The *z*-score for $x = -160.58$ is $z = -1.5$.
The *z*-score for $y = 162.85$ is $z = -1.5$.
Both $x = 160.58$ and $y = 162.85$ deviate the same number of standard deviations from their respective means and in the same direction.

TRY IT 6.4

In 2012, 1,664,479 students took the SAT exam. The distribution of scores in the verbal section of the SAT had a mean $\mu = 496$ and a standard deviation $\sigma = 114$. Let X = a SAT exam verbal section score in 2012. Then $X \sim N(496, 114)$.

Find the *z*-scores for $x_1 = 325$ and $x_2 = 366.21$. Interpret each *z*-score. What can you say about $x_1 = 325$ and $x_2 = 366.21$ as they compare to their respective means and standard deviations?

EXAMPLE 6.5

Suppose x has a normal distribution with mean 50 and standard deviation 6.

- About 68% of the x values lie within one standard deviation of the mean. Therefore, about 68% of the x values lie between $-1\sigma = (-1)(6) = -6$ and $1\sigma = (1)(6) = 6$ of the mean 50. The values $50 - 6 = 44$ and $50 + 6 = 56$ are within one standard deviation from the mean 50. The *z*-scores are –1 and +1 for 44 and 56, respectively.
- About 95% of the x values lie within two standard deviations of the mean. Therefore, about 95% of the x values lie between $-2\sigma = (-2)(6) = -12$ and $2\sigma = (2)(6) = 12$. The values $50 - 12 = 38$ and $50 + 12 = 62$ are within two standard deviations from the mean 50. The *z*-scores are –2 and +2 for 38 and 62, respectively.
- About 99.7% of the x values lie within three standard deviations of the mean. Therefore, about 99.7% of the x values lie between $-3\sigma = (-3)(6) = -18$ and $3\sigma = (3)(6) = 18$ from the mean 50. The values $50 - 18 = 32$ and $50 + 18 = 68$ are within three standard deviations of the mean 50. The *z*-scores are –3 and +3 for 32 and 68, respectively.

TRY IT 6.5

Suppose X has a normal distribution with mean 25 and standard deviation five. Between what values of x do 68% of

the values lie?

EXAMPLE 6.6

Problem

From 1984 to 1985, the mean height of 15 to 18-year-old males from Chile was 172.36 cm, and the standard deviation was 6.34 cm. Let Y = the height of 15 to 18-year-old males in 1984 to 1985. Then $Y \sim N(172.36, 6.34)$.

a. About 68% of the y values lie between what two values? These values are _____. The z-scores are _____, respectively.
b. About 95% of the y values lie between what two values? These values are _____. The z-scores are _____ respectively.
c. About 99.7% of the y values lie between what two values? These values are _____. The z-scores are _____, respectively.

Solution

a. About 68% of the values lie between 166.02 cm and 178.7 cm. The z-scores are –1 and 1.
b. About 95% of the values lie between 159.68 cm and 185.04 cm. The z-scores are –2 and 2.
c. About 99.7% of the values lie between 153.34 cm and 191.38 cm. The z-scores are –3 and 3.

TRY IT 6.6

The scores on a college entrance exam have an approximate normal distribution with mean, $\mu = 52$ points and a standard deviation, $\sigma = 11$ points.

a. About 68% of the y values lie between what two values? These values are _____. The z-scores are _____, respectively.
b. About 95% of the y values lie between what two values? These values are _____. The z-scores are _____, respectively.
c. About 99.7% of the y values lie between what two values? These values are _____. The z-scores are _____, respectively.

6.2 Using the Normal Distribution

The shaded area in the following graph indicates the area to the left of x. This area is represented by the probability $P(X < x)$. Normal tables, computers, and calculators provide or calculate the probability $P(X < x)$.

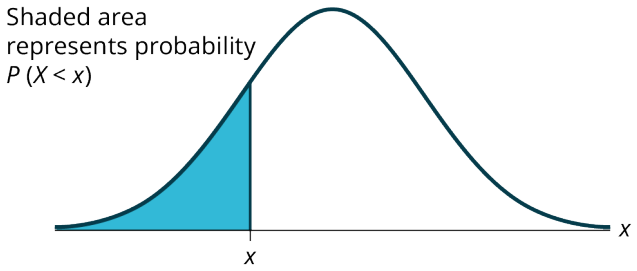


Figure 6.4

The area to the right is then $P(X > x) = 1 - P(X < x)$. Remember, $P(X < x)$ = **Area to the left** of the vertical line through x. $P(X > x) = 1 - P(X < x)$ = **Area to the right** of the vertical line through x. $P(X < x)$ is the same as $P(X \leq x)$ and $P(X > x)$ is the same as $P(X \geq x)$ for continuous distributions.

Calculations of Probabilities

Probabilities are calculated using technology. There are instructions given as necessary for the TI-83+ and TI-84 calculators.

NOTE

To calculate the probability, use the probability tables provided in Appendix H Tables without the use of technology. The tables include instructions for how to use them.

EXAMPLE 6.7

If the area to the left is 0.0228, then the area to the right is 1 – 0.0228 = 0.9772.

TRY IT 6.7

If the area to the left of x is 0.012, then what is the area to the right?

EXAMPLE 6.8

The final exam scores in a statistics class were normally distributed with a mean of 63 and a standard deviation of five.

Problem

a. Find the probability that a randomly selected student scored more than 65 on the exam.

Solution

a. Let X = a score on the final exam. $X \sim N(63, 5)$, where $\mu = 63$ and $\sigma = 5$.

Draw a graph.

Then, find $P(x > 65)$.

$P(x > 65) = 0.3446$

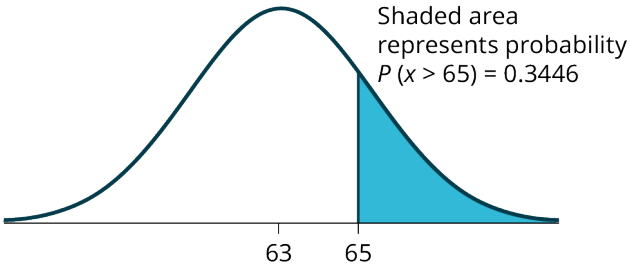


Figure 6.5

The probability that any student selected at random scores more than 65 is 0.3446.

USING THE TI-83, 83+, 84, 84+ CALCULATOR

Go into 2nd DISTR.
After pressing 2nd DISTR, press 2:normalcdf.

The syntax for the instructions are as follows:

normalcdf(lower value, upper value, mean, standard deviation) For this problem: normalcdf(65,1E99,63,5) = 0.3446. You get 1E99 (= 10^{99}) by pressing 1, the EE key (a 2nd key) and then 99. Or, you can enter 10^99 instead. The number 10^{99} is way out in the right tail of the normal curve. We are calculating the area between 65 and 10^{99}. In some

instances, the lower number of the area might be –1E99 (= -10^{99}). The number -10^{99} is way out in the left tail of the normal curve.

Historical Note

The TI probability program calculates a *z*-score and then the probability from the *z*-score. Before technology, the *z*-score was looked up in a standard normal probability table (because the math involved is too cumbersome) to find the probability. In this example, a standard normal table with area to the left of the *z*-score was used. You calculate the *z*-score and look up the area to the left. The probability is the area to the right.

$z = \frac{65 - 63}{5} = 0.4$

Area to the left is 0.6554.

$P(x > 65) = P(z > 0.4) = 1 - 0.6554 = 0.3446$

USING THE TI-83, 83+, 84, 84+ CALCULATOR

Find the percentile for a student scoring 65:

*Press 2nd Distr
*Press 2:normalcdf(
*Enter lower bound, upper bound, mean, standard deviation followed by)
*Press ENTER.
For this Example, the steps are
2nd Distr
2:normalcdf(65,1,2nd EE,99,63,5) ENTER

The probability that a selected student scored more than 65 is 0.3446.
To find the probability that a selected student scored *more than* 65, subtract the percentile from 1.

Problem

b. Find the probability that a randomly selected student scored less than 85.

Solution

b. Draw a graph.

Then find $P(x < 85)$, and shade the graph.

Using a computer or calculator, find $P(x < 85) = 1$.

normalcdf(0,85,63,5) = 1 (rounds to one)

The probability that one student scores less than 85 is approximately one (or 100%).

Problem

c. Find the 90th percentile (that is, find the score *k* that has 90% of the scores below *k* and 10% of the scores above *k*).

Solution

c. Find the 90th percentile. For each problem or part of a problem, draw a new graph. Draw the *x*-axis. Shade the area that corresponds to the 90th percentile.

Let *k* = the 90th percentile. The variable *k* is located on the *x*-axis. $P(x < k)$ is the area to the left of *k*. The 90th percentile *k* separates the exam scores into those that are the same or lower than *k* and those that are the same or higher. Ninety

percent of the test scores are the same or lower than k, and ten percent are the same or higher. The variable k is often called a **critical value**.

$k = 69.4$

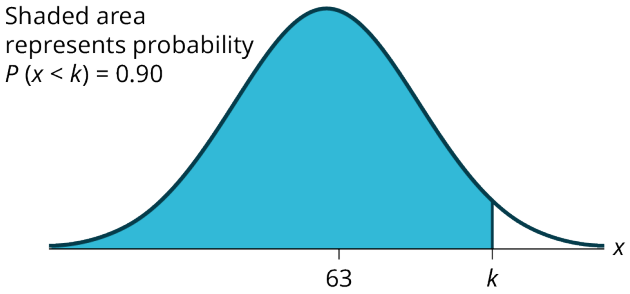


Figure 6.6

The 90th percentile is 69.4. This means that 90% of the test scores fall at or below 69.4 and 10% fall at or above. To get this answer on the calculator, follow this step:

USING THE TI-83, 83+, 84, 84+ CALCULATOR

`invNorm` in `2nd DISTR`. invNorm(area to the left, mean, standard deviation)
For this problem, invNorm(0.90,63,5) = 69.4

Problem

d. Find the 70th percentile (that is, find the score k such that 70% of scores are below k and 30% of the scores are above k).

Solution

d. Find the 70th percentile.

Draw a new graph and label it appropriately. $k = 65.6$

The 70th percentile is 65.6. This means that 70% of the test scores fall at or below 65.5 and 30% fall at or above.

invNorm(0.70,63,5) = 65.6

TRY IT 6.8

The golf scores for a school team were normally distributed with a mean of 68 and a standard deviation of three.

Find the probability that a randomly selected golfer scored less than 65.

EXAMPLE 6.9

A personal computer is used for office work at home, research, communication, personal finances, education, entertainment, social networking, and a myriad of other things. Suppose that the average number of hours a household personal computer is used for entertainment is two hours per day. Assume the times for entertainment are normally distributed and the standard deviation for the times is half an hour.

Problem

a. Find the probability that a household personal computer is used for entertainment between 1.8 and 2.75 hours per day.

Solution

a. Let X = the amount of time (in hours) a household personal computer is used for entertainment. $X \sim N(2, 0.5)$ where $\mu = 2$ and $\sigma = 0.5$.

Find $P(1.8 < x < 2.75)$.

The probability for which you are looking is the area **between** $x = 1.8$ and $x = 2.75$. $P(1.8 < x < 2.75) = 0.5886$

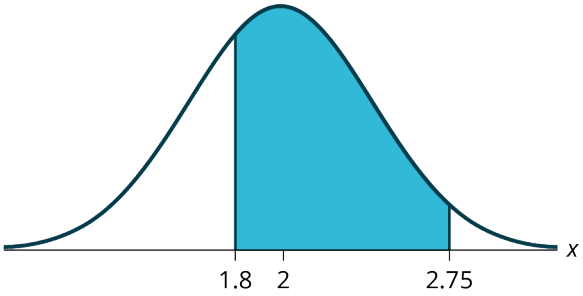


Figure 6.7

normalcdf(1.8,2.75,2,0.5) = 0.5886

The probability that a household personal computer is used between 1.8 and 2.75 hours per day for entertainment is 0.5886.

Problem

b. Find the maximum number of hours per day that the bottom quartile of households uses a personal computer for entertainment.

Solution

b. To find the maximum number of hours per day that the bottom quartile of households uses a personal computer for entertainment, **find the 25th percentile,** k, where $P(x < k) = 0.25$.

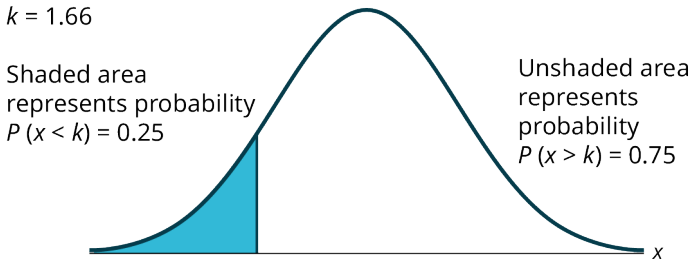


Figure 6.8

invNorm(0.25,2,0.5) = 1.66

The maximum number of hours per day that the bottom quartile of households uses a personal computer for entertainment is 1.66 hours.

TRY IT 6.9

The golf scores for a school team were normally distributed with a mean of 68 and a standard deviation of three. Find the probability that a golfer scored between 66 and 70.

EXAMPLE 6.10

In the United States the ages 13 to 55+ of smartphone users approximately follow a normal distribution with

approximate mean and standard deviation of 36.9 years and 13.9 years, respectively.

Problem

a. Determine the probability that a random smartphone user in the age range 13 to 55+ is between 23 and 64.7 years old.

b. Determine the probability that a randomly selected smartphone user in the age range 13 to 55+ is at most 50.8 years old.

c. Find the 80th percentile of this distribution, and interpret it in a complete sentence.

Solution

a. normalcdf(23,64.7,36.9,13.9) = 0.8186

b. normalcdf(-10^{99},50.8,36.9,13.9) = 0.8413

c.

invNorm(0.80,36.9,13.9) = 48.6
The 80th percentile is 48.6 years.
80% of the smartphone users in the age range 13 – 55+ are 48.6 years old or less.

TRY IT 6.10

Use the information in Example 6.10 to answer the following questions.

a. Find the 30th percentile, and interpret it in a complete sentence.
b. What is the probability that the age of a randomly selected smartphone user in the range 13 to 55+ is less than 27 years old.

EXAMPLE 6.11

In the United States the ages 13 to 55+ of smartphone users approximately follow a normal distribution with approximate mean and standard deviation of 36.9 years and 13.9 years respectively. Using this information, answer the following questions (round answers to one decimal place).

Problem

a. Calculate the interquartile range (*IQR*).

b. Forty percent of the smartphone users from 13 to 55+ are at least what age?

Solution

a.

$IQR = Q_3 - Q_1$
Calculate $Q_3 = 75^{th}$ percentile and $Q_1 = 25^{th}$ percentile.
invNorm(0.75,36.9,13.9) = Q_3 = 46.2754
invNorm(0.25,36.9,13.9) = Q_1 = 27.5246
$IQR = Q_3 - Q_1 = 18.8$

b.

Find k where $P(x \geq k) = 0.40$ ("At least" translates to "greater than or equal to.")
0.40 = the area to the right.
Area to the left = 1 – 0.40 = 0.60.
The area to the left of k = 0.60.

invNorm(0.60,36.9,13.9) = 40.4215.
k = 40.4.
Forty percent of the smartphone users from 13 to 55+ are at least 40.4 years.

TRY IT 6.11

Two thousand students took an exam. The scores on the exam have an approximate normal distribution with a mean μ = 81 points and standard deviation σ = 15 points.

a. Calculate the first- and third-quartile scores for this exam.
b. The middle 50% of the exam scores are between what two values?

EXAMPLE 6.12

A citrus farmer who grows mandarin oranges finds that the diameters of mandarin oranges harvested on his farm follow a normal distribution with a mean diameter of 5.85 cm and a standard deviation of 0.24 cm.

Problem

a. Find the probability that a randomly selected mandarin orange from this farm has a diameter larger than 6.0 cm. Sketch the graph.

b. The middle 20% of mandarin oranges from this farm have diameters between _____ and _____.

c. Find the 90th percentile for the diameters of mandarin oranges, and interpret it in a complete sentence.

Solution

a. normalcdf(6,10^99,5.85,0.24) = 0.2660

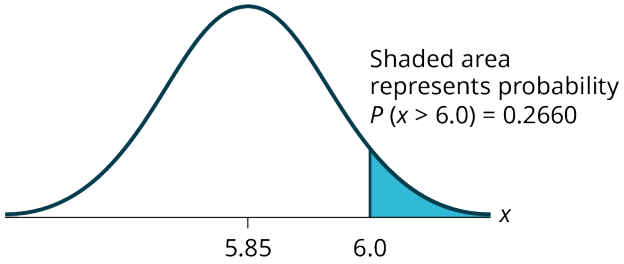


Figure 6.9

b.

1 – 0.20 = 0.80
The tails of the graph of the normal distribution each have an area of 0.40.
Find $k1$, the 40th percentile, and $k2$, the 60th percentile (0.40 + 0.20 = 0.60).
$k1$ = invNorm(0.40,5.85,0.24) = 5.79 cm
$k2$ = invNorm(0.60,5.85,0.24) = 5.91 cm

c. 6.16: Ninety percent of the diameter of the mandarin oranges is at most 6.16 cm.

TRY IT 6.12

Using the information from Example 6.12, answer the following: